A dimension-independent box bound from dual total correlation

Sam Vaseghi*

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Abstract

Let S be a finite nonempty set in a product of d coordinate spaces, and let U_S be its uniform law. We prove that some nonempty Cartesian box B satisfies

$$|S \triangle B| \le 10\,\mathrm{DTC}(\mathrm{U}_S)|S|,$$

where dual total correlation is computed using natural logarithms. The constant does not depend on the dimension or on the coordinate alphabets. The proof follows actual coordinate slices: their average disagreement is bounded linearly by mutual information, while an exact entropy identity retains the deficit available within each slice. Selecting one slice with both costs controlled makes the induction close with the same constant. The resulting Loomis–Whitney stability estimate has coefficient $20(d-1)$ under the usual normalised near-equality assumption. A two-point example shows that linear dependence on d is necessary in that normalisation. The constants are not claimed sharp.

Keywords: dual total correlation; Loomis–Whitney inequality; stability; Cartesian boxes; entropy inequalities.

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1 From nearly sharp projections to a box

A finite Cartesian box consists of all combinations of its allowed coordinate values. If $B = B_1 \times \cdots \times B_d$, deleting coordinate i leaves a projection of size $\prod_{k \ne i} |B_k|$. Each factor $|B_k|$ occurs exactly $d-1$ times in the product of these projection sizes, and hence

$$\prod_{i=1}^{d} |\pi_{-i} B| = |B|^{d-1}.$$

For an arbitrary finite set S, the Loomis–Whitney inequality gives $|S|^{d-1} \le \prod_i |\pi_{-i} S|$. Its stability problem asks whether a small discrepancy in this inequality forces S to differ from some box in only a small fraction of its points.

Ellis, Friedgut, Kindler and Yehudayoff established such stability through information-theoretic arguments [4]. Their Corollary 3 gives the coefficient $(d-1)(4d^2+64d)$ in the normalisation used below. Their work connects entropy deficit with geometric reconstruction, a connection also discussed by Austin in terms of dual total correlation [1, Section 7.3]. Earlier geometric refinements for near equality in the Loomis–Whitney inequality were obtained by Zhang [5]. The question addressed here is the dependence of the reconstruction error on dimension: we obtain an absolute linear coefficient in terms of dual total correlation, and consequently a coefficient of linear order in d for the normalised projection deficit.

*Wisehouse Science Communication Lab, Wisehouse AB, Havrabjärsgatan 6, SE-28197 Ballingslöv, Sweden. Email: sam.vaseghi@wisehouse-publishing.com. ORCID: 0009-0006-4105-1949.

Let $X = (X_1, \dots, X_d)$ have finite support. We use natural logarithms and write

$$\mathrm{DTC}(X) = \sum_{i=1}^{d} H(X_{-i}) - (d-1)H(X), \tag{1}$$

where X_{-i} is the tuple obtained by deleting X_i, and H is Shannon entropy. An empty tuple has entropy zero, so $\mathrm{DTC}(X_1) = 0$. For a finite nonempty set S, the notation $\mathrm{DTC}(\mathrm{U}_S)$ means the value in (1) when X is uniform on S. Then $H(X) = \log|S|$, while $H(X_{-i}) \le \log|\pi_{-i}S|$. Thus DTC is at most the logarithmic projection deficit $\sum_i \log|\pi_{-i}S| - (d-1)\log|S|$: nearly sharp projections give a small DTC, and a reconstruction bound in terms of DTC will imply projection stability.

The distance used here is relative symmetric difference. Thus $|S \triangle B|/|S|$ counts both the points removed from S and the points added to obtain B, normalised by the original size of S.

Theorem 1. *For every integer $d \ge 1$, every collection of coordinate sets $\Omega_1, \dots, \Omega_d$, and every finite nonempty set $S \subseteq \prod_{i=1}^{d} \Omega_i$, there are finite nonempty sets $B_i \subseteq \Omega_i$ such that, for $B = \prod_i B_i$,*

$$\frac{|S \triangle B|}{|S|} \le 10\,\mathrm{DTC}(\mathrm{U}_S). \tag{2}$$

Only the coordinates occurring in S are used; the ambient sets need not be finite. The hypothesis that the law is uniform on a finite support will enter at the distance estimate in Section 3.

Corollary 2. *Let $d \ge 2$ and let S satisfy the hypotheses of Theorem 1. If $0 \le \varepsilon < 1$ and*

$$|S| \ge (1-\varepsilon)\left(\prod_{i=1}^{d} |\pi_{-i}S|\right)^{1/(d-1)}, \tag{3}$$

then some finite nonempty box B satisfies

$$\frac{|S \triangle B|}{|S|} \le 20(d-1)\varepsilon. \tag{4}$$

Corollary 2 establishes the Loomis–Whitney specialization of Conjecture 20 of [4]. Section 7 gives the parameter comparison and distinguishes this case from the general uniform-cover conjecture.

We first prove the entropy statement and then deduce the projection estimate. The proof reconstructs S in two stages. It replaces the slices perpendicular to the last coordinate by copies of one actual slice, forming a cylinder, and then replaces that reference slice by a box in one fewer dimension. The choice of reference slice must control both replacements. Keeping that choice inside the original family of slices allows an exact entropy identity to account for the second cost.

2 The entropy retained inside the slices

For a finite law Q, entropy is $H(Q) = -\sum_x Q(x)\log Q(x)$, with $0\log 0 = 0$. Conditional entropy is the average entropy of the conditional laws. Relative entropy and mutual information are

$$D(Q\|P) = \sum_{x:Q(x)>0} Q(x)\log\frac{Q(x)}{P(x)}, \qquad I(Y;Z) = D(P_{YZ}\|P_Y \otimes P_Z).$$

The relative entropy is infinite if P vanishes where Q is positive. The identity $I(Y;Z) = H(Y) + H(Z) - H(Y,Z)$ and the chain rule $H(Y,Z) = H(Z) + H(Y \mid Z)$ follow by expanding the finite sums.

For $d \ge 2$, split X as (Y,Z), where $Y = (X_1, \dots, X_{d-1})$ and $Z = X_d$. Dual total correlation in the first $d-1$ coordinates, conditioned on Z, is

$$\mathrm{DTC}(Y \mid Z) = \sum_{i=1}^{d-1} H(Y_{-i} \mid Z) - (d-2)H(Y \mid Z).$$

Substituting $H(Y_{-i} \mid Z) = H(X_{-i}) - H(Z)$ gives

$$\mathrm{DTC}(Y \mid Z) = \sum_{i=1}^{d-1} H(X_{-i}) - (d-2)H(X) - H(Z).$$

Adding $I(Y;Z) = H(Y) + H(Z) - H(X)$ recovers (1), since $Y = X_{-d}$. We have proved the identity

$$\mathrm{DTC}(X) = I(Y;Z) + \mathrm{DTC}(Y \mid Z). \tag{5}$$

This is the DTC chain rule; see also Austin [2, Corollary 6.6]. For $d = 2$, the conditional DTC is zero, consistently with $\mathrm{DTC}(X_1, X_2) = I(X_1; X_2)$. Relative entropy is nonnegative, so induction in (5) also proves $\mathrm{DTC}(X) \geq 0$ in every dimension.

Now let X be uniform on S. Denote the last-coordinate values occurring in S by J, and write

$$S_j = \{y : (y, j) \in S\}, \qquad n = |S|, \quad s_j = |S_j|, \quad p_j = \frac{s_j}{n}, \quad \mathrm{U}_j = \mathrm{U}_{S_j}. \tag{6}$$

Every S_j is nonempty, $p_j > 0$, and $\sum_j p_j = 1$. The conditional law of Y given $Z = j$ is exactly U_j, while the unconditional law of Y is the mixture

$$P = \sum_{j \in J} p_j \mathrm{U}_j. \tag{7}$$

This marginal need not be uniform on the union of the slices: points appearing in several slices receive more mass.

Put $\delta = \mathrm{DTC}(\mathrm{U}_S)$, $\delta_j = \mathrm{DTC}(\mathrm{U}_j)$ and $I = I(Y;Z)$. Identity (5) becomes

$$\delta = I + \sum_j p_j \delta_j. \tag{8}$$

The first term measures dependence between the slice label and the point in the slice. The second retains the average deficit within the slices, where the induction will act. To use the first term for reconstruction, we next turn it into a bound on how much two slice laws can disagree.

3 Uniform support converts information into set disagreement

We use total variation with the convention $\mathrm{TV}(Q, R) = \frac{1}{2}\sum_x |Q(x) - R(x)|$. For two finite nonempty sets E, F, let $e = |E|$, $f = |F|$ and $c = |E \cap F|$. The common mass of their uniform laws is $c/\max(e, f)$, and consequently

$$\mathrm{TV}(\mathrm{U}_E, \mathrm{U}_F) = 1 - \frac{c}{\max(e, f)} = \max\{\mathrm{U}_E(F^c), \mathrm{U}_F(E^c)\}. \tag{9}$$

This formula controls total variation by mass lying outside the other support. It uses uniformity of both laws; matching supports alone would give no such control for arbitrary probability weights.

We also need a support consequence of relative entropy. If Q is supported on E and $P(E) > 0$, then

$$D(Q\|P) = D\big(Q\|P(\cdot \mid E)\big) - \log P(E) \geq -\log P(E) \geq 1 - P(E). \tag{10}$$

If the first relative entropy is infinite, the inequality remains valid. In our application it is finite because the mixture (7) gives positive mass at every point of every slice. Estimate (10) is the same support-hole principle used in the information-theoretic approach of [4].

For a possible reference slice j, define its average disagreement with the other slices by

$$v_j = \sum_{k \in J} p_k \,\mathrm{TV}(\mathrm{U}_j, \mathrm{U}_k). \tag{11}$$

The average over reference choices is controlled by I.

Lemma 3. *With the notation above,*

$$\sum_j p_j v_j \leq 2I. \tag{12}$$

Proof. By (9), total variation is at most the sum of the two outside-support masses. Interchanging j and k makes their weighted double sums equal. Thus

$$\begin{aligned}\sum_j p_j v_j &\le \sum_{j,k} p_j p_k \big[\mathrm{U}_j(S_k^c) + \mathrm{U}_k(S_j^c)\big] \\ &= 2\sum_j p_j P(S_j^c) \\ &\le 2\sum_j p_j D(\mathrm{U}_j \| P) = 2I.\end{aligned}$$

The last equality is the conditional-law expression for mutual information. □

The linear dependence in (12) comes from comparing uniform supports and measuring their missing mass. A general total-variation estimate from relative entropy would normally introduce a square root. Here the uniform conditional laws let us retain the linear scale needed for the final cardinality bound.

4 Replacing the slices by one cylinder

Fix a reference index j for the moment, and put $R = S_j$ and $r = |R|$. We retain those slice labels whose slices are at least half as large as R:

$$A_j = \{k \in J : s_k \ge r/2\}. \tag{13}$$

The reference label belongs to A_j, so the cylinder $R \times A_j$ is nonempty. We do not require its first factor R to be a box yet.

The threshold has two uses. On a retained slice, $r \le 2s_k$, which bounds the cost of replacing S_k by R in units of the original slice size. On a discarded slice, uniformity forces a total-variation discrepancy exceeding $1/2$, which pays for deleting that slice. Figure 1 shows these set operations for a chosen reference slice.

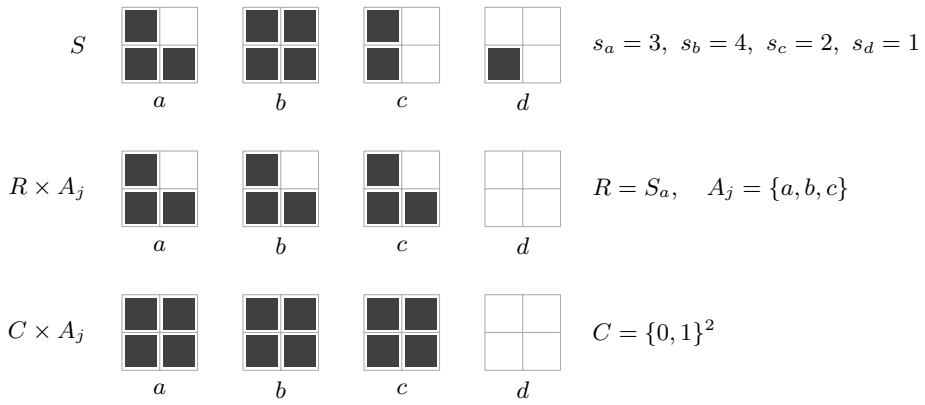


Figure 1: Each filled cell represents a point in a two-coordinate slice; the column labels are values of the third coordinate. Taking the first slice as reference gives $r = 3$ and retains the slices of sizes $3, 4, 2$. Repeating R on those labels forms the middle cylinder. Completing R to the square C gives the bottom box. This finite example illustrates the two replacements for a chosen reference; it is not an instance of the small-deficit selection used in the proof.

Lemma 4. *For every reference slice j, the cylinder from* (13) *satisfies*

$$\frac{|S \triangle (R \times A_j)|}{n} \le 3v_j, \qquad \frac{r|A_j|}{n} \le 1 + 2v_j. \tag{14}$$

Proof. First, (9) implies for any finite nonempty E, F that

$$|E \triangle F| \le (|E| + |F|)\,\mathrm{TV}(\mathrm{U}_E, \mathrm{U}_F). \tag{15}$$

Indeed, with $e = |E|$, $f = |F|$, $c = |E \cap F|$ and $m = \max(e, f)$, the right side minus the left side is $c(2 - (e + f)/m) \geq 0$.

Put $t_k = \mathrm{TV}(\mathrm{U}_R, \mathrm{U}_k)$. If $k \in A_j$, then $r \leq 2s_k$, so

$$|S_k \triangle R| \leq (s_k + r)t_k \leq 3s_k t_k.$$

If $k \notin A_j$, then $s_k < r/2$ and

$$t_k = 1 - \frac{|S_k \cap R|}{r} \geq 1 - \frac{s_k}{r} > \frac{1}{2}.$$

Deleting this slice costs $s_k \leq 2s_k t_k$. The fibres at distinct labels are disjoint, hence

$$|S \triangle (R \times A_j)| = \sum_{k \in A_j} |S_k \triangle R| + \sum_{k \notin A_j} s_k \leq 3 \sum_k s_k t_k = 3nv_j.$$

We also need to control the number of copies of R, because replacing R by a smaller-dimensional box will be repeated on every retained label. For $k \in A_j$, the bound $r \leq s_k + 2s_k t_k$ is immediate if $r \leq s_k$. If $r > s_k$, then

$$r - s_k = r(1 - s_k/r) \leq r t_k \leq 2s_k t_k.$$

Summing over retained labels gives

$$r|A_j| \leq \sum_{k \in A_j} s_k + 2 \sum_{k \in A_j} s_k t_k \leq n(1 + 2v_j),$$

which proves the second estimate. Equality at the threshold $s_k = r/2$ is included among the retained slices and causes no change in the argument. □

The first estimate bounds the immediate replacement error. The second controls how much the inductive error inside R will grow when copied across A_j. Both must enter the choice of reference slice; controlling the first alone would not account for the final box.

5 Choosing the slice and completing the induction

Proof of Theorem 1. We induct on d. In dimension one, S itself is a box and its DTC is zero. Suppose $d \geq 2$ and the theorem is known in dimension $d - 1$.

If $\delta \geq 1/10$, choose any $x \in S$ and take the singleton box $B = \prod_i \{x_i\}$. Then

$$\frac{|S \triangle B|}{n} = 1 - \frac{1}{n} \leq 1 \leq 10\delta.$$

We may therefore assume $\delta < 1/10$.

The reference slice should have small average disagreement v_j and small internal deficit δ_j. Instead of choosing for these requirements separately, combine them before averaging. By (8) and Lemma 3,

$$\sum_j p_j(5v_j + 10\delta_j) \leq 10I + 10 \sum_j p_j \delta_j = 10\delta. \tag{16}$$

There is consequently an index j with

$$5v_j + 10\delta_j \leq 10\delta < 1. \tag{17}$$

Fix that index, put $R = S_j$, $r = |R|$ and $a = 10\delta_j$, and use its set A_j from (13). In particular $0 \leq a < 1$.

The inductive hypothesis applied to the actual slice R supplies a nonempty box C in the first $d - 1$ coordinates with $|R \triangle C| \leq ar$. Set $B = C \times A_j$. The symmetric-difference triangle inequality and the fact that every discrepancy between R and C is copied $|A_j|$ times give

$$\begin{aligned} \frac{|S \triangle B|}{n} &\leq \frac{|S \triangle (R \times A_j)|}{n} + \frac{|(R \times A_j) \triangle (C \times A_j)|}{n} \\ &= \frac{|S \triangle (R \times A_j)|}{n} + \frac{|A_j|\,|R \triangle C|}{n} \\ &\leq 3v_j + (1 + 2v_j)a \\ &\leq 5v_j + a \leq 10\delta. \end{aligned} \tag{18}$$

Here the two cylinder bounds control the third line, and $a \le 1$ gives the next inequality. The last inequality is the joint selection (17). Both factors of B are nonempty, and the induction is complete. □

The coefficient 5 in the selection is accounted for by the last calculation: $3v_j$ pays for replacing S by the cylinder, and at most $2v_j$ pays for copying the inductive error. Averaging $5v_j$ then costs at most $10I$, matching the coefficient 10 already assigned to the conditional deficit. This matching prevents a dimension factor from accumulating at each induction step.

No limiting argument is needed at zero deficit. When $\delta = 0$, the selected v_j and δ_j both vanish, and the constructed box equals S. Empty slices never enter the family J, so no uniform law on an empty set is used.

6 The projection consequence and its dimension dependence

Proof of Corollary 2. For $X \sim \mathrm{U}_S$, we have $H(X) = \log|S|$, and each X_{-i} is supported on $\pi_{-i}S$. Entropy is at most the logarithm of support size, so

$$\mathrm{DTC}(\mathrm{U}_S) \le \Delta_{\mathrm{LW}}(S) := \sum_i \log|\pi_{-i}S| - (d-1)\log|S|. \tag{19}$$

Taking logarithms in (3) gives $\Delta_{\mathrm{LW}}(S) \le -(d-1)\log(1-\varepsilon)$. If $0 \le \varepsilon \le 1/2$, then $-\log(1-\varepsilon) \le 2\varepsilon$, and Theorem 1 proves (4). If $\varepsilon > 1/2$, any singleton box inside S has relative error at most one, which is less than $20(d-1)\varepsilon$. □

The same argument gives the more precise logarithmic estimate

$$\frac{|S \triangle B|}{|S|} \le -10(d-1)\log(1-\varepsilon).$$

The linear form (4) makes comparison with the usual stability normalisation immediate. Its order in d cannot be reduced to an absolute coefficient.

Proposition 5. *For $d \ge 2$, let $S = \{0^d, 1^d\} \subseteq \{0,1\}^d$. Then the minimum relative error over nonempty boxes is $1/2$, while (3) holds with equality for*

$$\varepsilon_d = 1 - 2^{-1/(d-1)} \sim \frac{\log 2}{d-1}.$$

Proof. Every coordinate-deletion projection has size two, giving the stated ε_d. A singleton at either point of S has relative error $1/2$. A box containing exactly one point of S has at least one error, and a box containing neither has more. A box containing both must contain every combination of zeros and ones, so it has 2^d points and at least $2^d - 2 \ge 2$ errors. Thus the singleton value is minimal. Finally, expand $1 - e^{-\log 2/(d-1)}$ as $d \to \infty$. □

If a stability bound in this normalisation has coefficient c_d, this example forces $c_d \ge [2\varepsilon_d]^{-1} \sim (d-1)/(2\log 2)$. Corollary 2 therefore has the necessary order of dimension dependence. This says nothing about the best numerical constant in either theorem.

7 Relation to earlier stability bounds

The entropy-to-box strategy, conditioning identities and support-hole estimates used above have precedents in [4, 1, 2]. To compare the dimension dependence directly in the entropy formulation, Appendix A derives from the trimming argument of [4] the estimate

$$\inf_{B \text{ nonempty box}} \frac{|S \triangle B|}{|S|} \le (2d^2 + 32d)\,\mathrm{DTC}(\mathrm{U}_S). \tag{20}$$

This is a consequence we extract from that proof, rather than a theorem stated there in this form. Theorem 1 replaces this dimension-dependent coefficient by 10. The change in the construction is

to keep the reference factor as an actual conditional slice. Its deficit then remains part of the exact identity (8), and the joint selection (17) pays for the successive set replacements without introducing a new dimension factor.

Bennett and Oh [3] establish stability for the functional Finner inequality and its adjoint on product measure spaces. Their set consequence gives a linear bound in the normalised projection deficit with an unspecified multiplicative constant [3, Remark 1.3 and Corollary 2.4], while their adjoint stability theorem gives an ε–δ conclusion for fixed $0 < p < 1$ [3, Theorem 1.4]. They also record the Shannon limit of the underlying Rényi entropy inequality [3, Remark 1.6], without stating a dimension-uniform stability modulus for that limit. Theorem 1 provides an explicit dimension-independent linear bound in DTC for uniform laws on finite sets; obtaining this conclusion from their results would require further quantitative arguments.

For the coordinate-complement cover, the parameters in Conjecture 20 of [4] are $m = d - 1$ and $\sigma = 1$. Here m counts the cover sets containing each coordinate, and σ is the minimum, over distinct coordinates i, j, of the number of cover sets containing i but not j. Corollary 2 establishes this Loomis–Whitney specialization with coefficient $20m/\sigma = 20(d-1)$. The conjecture for general uniform covers remains outside the scope of the present argument. Proposition 5 shows that the order of dimension dependence in the specialization is necessary; the best numerical constants in Theorem 1 and Corollary 2 remain undetermined.

A A DTC bound extracted from the earlier trimming argument

We give the deduction of (20) so that the comparison can be checked in the same entropy normalisation as Theorem 1. The only additional ingredient is the marginal trimming statement of Ellis, Friedgut, Kindler and Yehudayoff [4, Lemma 12]. Their entropy convention uses base-two logarithms; the support estimates below are instead derived from (10), with natural logarithms throughout.

Let $d \geq 2$, $p = \mathrm{U}_S$, $n = |S|$, $\delta = \mathrm{DTC}(p)$ and $K_d = 2d^2 + 32d$. If $\delta \geq 1/K_d$, a singleton in S proves the bound. Assume therefore that $\delta < 1/K_d$, and restrict each coordinate space to the finite projection of S. Write p_i, p_{-i} for the marginals and put

$$h_i = (p_i \otimes p_{-i})(S^c), \qquad q = \bigotimes_{i=1}^{d} p_i, \qquad h = q(S^c).$$

Applying (5) with coordinate i singled out gives $I(X_i; X_{-i}) \leq \delta$. The chain rule for mutual information and its monotonicity under enlarging the second tuple give

$$D(p\|q) = \sum_{i=2}^{d} I(X_i; X_1, \ldots, X_{i-1}) \leq \sum_{i=2}^{d} I(X_i; X_{-i}) \leq (d-1)\delta.$$

Since p is supported on S, (10) applied with reference laws $p_i \otimes p_{-i}$ and q now yields

$$h_i \leq \delta, \qquad h \leq (d-1)\delta. \tag{21}$$

View S as a two-block set with coordinate i in one block and all remaining coordinates in the other. Lemma 12 of [4], with its parameter $\alpha = 1/d$, supplies a coordinate set R_i such that

$$p_i(R_i^c) \leq 2dh_i, \qquad p_i(x_i) \geq \frac{(1 - 2dh_i)(1 - 1/d)}{|R_i|} \quad (x_i \in R_i).$$

Our smallness assumption gives $2d\delta < 1/(d+16) < 1/d$. Thus the retained marginal mass is positive, every R_i is nonempty, and the last lower bound is at least $(1-1/d)^2/|R_i|$. Set $R = \prod_i R_i$. The union bound gives

$$\frac{|S \setminus R|}{n} = p(R^c) \leq 2d^2\delta. \tag{22}$$

For every $x \in R$, the product marginal satisfies

$$q(x) \geq \frac{(1 - 1/d)^{2d}}{|R|} \geq \frac{1}{16|R|},$$

using $(1-1/d)^d \geq 1/4$ for integers $d \geq 2$. Summing over $R \setminus S$ and using (21) shows

$$|R \setminus S| \leq 16d\delta|R|, \qquad n \geq (1-16d\delta)|R|.$$

Also $16d\delta < 8/(d+16) \leq 4/9 < 1/2$, so

$$\frac{|R \setminus S|}{n} \leq \frac{16d\delta}{1-16d\delta} \leq 32d\delta.$$

Adding (22) proves (20). In dimension one the bound holds with $R = S$.

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References

[1] T. Austin, *Measure concentration and the weak Pinsker property*, Publications Mathématiques de l'IHÉS **128** (2018), 1–119. https://doi.org/10.1007/s10240-018-0098-3. arXiv:1705.00302v4.

[2] T. Austin, *Multi-variate correlation and mixtures of product measures*, Kybernetika **56**(3) (2020), 459–499. https://doi.org/10.14736/kyb-2020-3-0459. arXiv:1809.10272v4.

[3] J. Bennett and S. Oh, *Stability for the adjoint Finner inequality and improved lower bounds on the k-plane transform*, arXiv preprint (2026), version 1, 21 October 2026. arXiv:2610.29041v1.

[4] D. Ellis, E. Friedgut, G. Kindler and A. Yehudayoff, *Geometric stability via information theory*, Discrete Analysis (2016), Paper No. 10, 28 pp. https://doi.org/10.19086/da.784. arXiv:1510.00258v4.

[5] R. Zhang, *On configurations where the Loomis–Whitney inequality is nearly sharp and applications to the Furstenberg set problem*, Mathematika **61**(1) (2015), 145–161. https://doi.org/10.1112/S0025579314000357. arXiv:1309.2372v2.